Design a seven-segment code

Task 4.1 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Lay out a figure 8 with A across the top, B upper right, C lower right, D along the bottom, E lower left, F upper left and G across the middle. Agree on 1 = lit and 0 = dark. Draw the switched lamp loop.

Mark the segment positions A–G in the same order on two cards. Count which segments light for one digit.

Digit	A B C D E F G	Digit	A B C D E F G
0		5	
1		6	
2		7	
3		8	
4		9	

02 Plan it on paper

Make an on/off code for digits 0–9. Agree on the bit order and the meaning of 1 and 0.

My prediction, chosen feature or design limit

Design a seven-segment code

Task 4.1 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Build a switched lamp circuit. One partner sends three secret digit patterns and the other reconstructs the display. Swap roles and decode three new patterns. If a digit is wrong, compare segment order before changing the key.

Round	Sent digit / bits	Decoded digit	Match or correction
A1			
A2			
A3			
B1			
B2			
B3			

04 Use the reading

Ch. 24 | A Highway for Ideas | printed pp. 221-230
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Show the code for one digit and explain how another person can decode it without seeing your drawing.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Test a detector for both missed and false signals

Task 4.2 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Place a paper target where the switch should trigger. Draw the source, trigger and output. Define what counts as a detection before testing.

Set the trigger location and record how many trials you will run with and without movement.

Condition	Before trials 1 / 2 / 3	After trials 1 / 2 / 3
Movement		
No movement		

02 Plan it on paper

Draw the source, trigger, and output. State what should happen in each condition.

My prediction, chosen feature or design limit

Test a detector for both missed and false signals

Task 4.2 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Assemble the detector and move the simulated trigger the same way for each trial. Record the output. Run three trigger and three no-trigger trials. Change one unreliable contact. Repeat both sets.

Version	True detections / 3	False alarms / 3	Contact change or note
Before			
After			

04 Use the reading

Ch. 25 | Teaching a Machine to Listen | printed pp. 231-236
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Give the detector’s true detections and false alarms. Which change would improve reliability?

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Send a message across a hidden channel

Task 4.3 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Draw transmitter, channel and receiver. Mark a conversion, the separation, the message key and start/stop rule. Keep the secret message out of sight.

Measure or mark a fixed separation. Agree on a small message set and the start/stop signal.

Signal or pattern	Meaning	Start / stop or timing rule
Message 1		
Message 2		
Message 3		

02 Plan it on paper

Draw the sender, channel, and receiver. Mark the point where the message changes form.

My prediction, chosen feature or design limit

Send a message across a hidden channel

Task 4.3 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Send messages chosen out of sight of the receiver. Use the audio or optical stages of the available communication kit. A coded lamp is a classroom substitute. Record decoded messages out of total attempts. Compare one changed condition at the same distance.

Run	Sent	Decoded	Condition / result
1			
2			
3			
Changed 1			
Changed 2			
Changed 3			

04 Use the reading

Ch. 25 | Teaching a Machine to Listen | printed pp. 231-236
Ch. 26 | Shedding Light on Communications | printed pp. 237-244
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Where did information change form? Use a decoded message or error to support your explanation.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

PROJECT 4 / MEETING 04 / COMMUNICATION AND CONTROL

Connect a lunar base to three destinations

Task 4.3 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Plan two-way links inside a habitat, from habitat to surface crew and from base to Earth. Sound in air needs a material medium. An electromagnetic signal can cross a vacuum. Draw conversions and a return path.

Mark sender, receiver, and the medium for each required route. Compare indoor, surface, and long-distance links.

Route	Medium / channel	Reason or limitation
Inside base		
Outside crew		
Earth link		

02 Plan it on paper

Choose a channel for each route. Give one reason related to environment, range, or obstruction.

My prediction, chosen feature or design limit

Connect a lunar base to three destinations

Task 4.3 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Create the system diagram and label conversions. Demonstrate an Earth-classroom model of one message path where equipment permits. Ask another team to follow a message from origin to destination. Identify an obstruction, failed component, or missing return path.

Review	Message path or fault	Test result or revision
Trace outward		
Trace return		
One obstruction		

04 Use the reading

Ch. 26 | Shedding Light on Communications | printed pp. 237-244
Ch. 27 | Riding the Waves | printed pp. 245-252
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Defend one channel choice and explain a limit of your classroom test for the lunar setting.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Measure current in one branch

Task 4.4 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Disconnect power before opening a low-voltage loop. Put the configured ammeter in series at the break. Have the teacher check the range, port and wiring before power is applied.

Trace the closed lamp path. Identify where you will break the branch and what unit you expect.

Meter location	Range / units	Expected reading	Teacher check
Point A			
Point B			

02 Plan it on paper

Draw the meter in series. Have a partner and teacher check the drawing and connection before energizing.

My prediction, chosen feature or design limit

Measure current in one branch

Task 4.4 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Take a current reading. Switch power off, move the meter to a second point in the same unbranched loop, then measure again. Compare readings and their precision. Investigate a large difference by checking range, battery and contacts.

Location	Reading and unit	Precision	Comparison or contact check
A			
B			
Repeat			

04 Use the reading

Ch. 28 | Designed Learning | printed pp. 253-262

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Where did you insert the meter and why? Report two readings and what the comparison shows.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Measure voltage between two points

Task 4.5 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Label node pairs and lead colors. Place the DC voltmeter across each selected pair. Record source voltage, lamp drop and the sign when leads are reversed.

Record the source voltage across its terminals. Label measurement points on a schematic.

Pair of nodes	Red lead at	Black lead at	Prediction and unit
Source			
Lamp			
Closed connection			

02 Plan it on paper

Predict the voltage across a lamp and across a closed connection. Mark both meter leads for each prediction.

My prediction, chosen feature or design limit

Measure voltage between two points

Task 4.5 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Connect the voltmeter in parallel at each pair of points. Record the setting, sign, and units. Reverse the leads on one measurement. Explain the sign change and compare the voltage drops round a closed loop.

Pair / condition	DC range	Signed reading (V)	Explanation
Source			
Lamp			
Connection			
Reversed pair			

04 Use the reading

Ch. 29 | On the Grid | printed pp. 263-270
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Show one lead placement and explain what the number measures between those two points.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Build the fan controls one state at a time

Task 4.6 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Label motor terminals and sketch the teacher-checked switch paths. Show where polarity reverses and what changes speed. Disconnect power before rewiring.

List the required speeds and directions in a state table. Mark the intended polarity at the motor for each direction.

State	Switch settings	Expected polarity	Predicted motion
Off			
Forward low			
Forward high			
Reverse low			
Reverse high			

02 Plan it on paper

Draw a circuit for each switch state. Trace the current path and identify the component changing the speed.

My prediction, chosen feature or design limit

Build the fan controls one state at a time

Task 4.6 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Use the teacher-checked diagram for the available kit. Build one state at a time. Verify direction and two distinct speed settings before combining the controls. Ask a partner to predict the motor state from the diagram before operating it. Record differences and disconnect power before correcting a connection.

State tested	Actual direction / speed	Mismatch or fix
Forward settings		
Reverse settings		
Off		

04 Use the reading

Ch. 30 | Electrifying! | printed pp. 271-280
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Which control changes direction and which changes speed in your circuit? Support it with an observation.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Commission a fan another crew can operate

Task 4.6 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Write the stop rule and define how you will distinguish direction and speed. Use the same observation method in every trial. Mark the revised control label on the sketch.

Mark a safe stop position and identify how to tell forward from reverse. Use the same method to compare speeds.

Control label	Expected state	Actual state	Retest result
Off			
Forward low			
Forward high			
Reverse low			
Reverse high			

02 Plan it on paper

Write a test for every required control state. Include a safe power-off check.

My prediction, chosen feature or design limit

Commission a fan another crew can operate

Task 4.6 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

One partner operates the controls from the labels while another reads the schematic. Exchange roles. Repair one fault or labeling problem. Retest both speeds and reversal. Give the control sheet to a neighboring crew and ask them to reproduce a selected state.

Verification	Operator instruction	Independent crew result
Safe stop		
Selected state		
Repaired fault		

04 Use the reading

Ch. 30 | Electrifying! | printed pp. 271-280
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Name the failed or unclear state, the correction, and the observed outcome after retesting.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Supply an island lighthouse for a full night

Task 4.7 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

The island lamp draws 1,000 W. Assume 12 h of darkness first. Compare two of these options: delivered batteries, solar with storage, wind with storage or a fuel generator. Consider service visits and a longer night.

Compute daily demand as lamp power × operating hours. Example: 10 W for 12 h uses 120 Wh. Keep W and Wh distinct.

Case	Lamp power (W)	Hours	Demand (Wh and kWh)
12-hour night			
Longer night			

02 Plan it on paper

Compare two candidate supplies. Include operating hours, storage, site conditions, and an assumption.

My prediction, chosen feature or design limit

Supply an island lighthouse for a full night

Task 4.7 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Prepare a proposal with a labeled energy-flow diagram, demand estimate, and a plan for hours when the source is unavailable. Have a partner change one condition such as a longer night. Recalculate and revise the recommendation.

Candidate	Supply and storage	Service or site limit	How it covers darkness
Option A			
Option B			

04 Use the reading

Ch. 31 | Sunny Side Up | printed pp. 281-287

Ch. 32 | Cape Wind | printed pp. 289-294

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

What source and storage plan would you recommend? Cite one calculated demand and one site constraint.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Repair and repeat a fair test

Flexible clinic / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

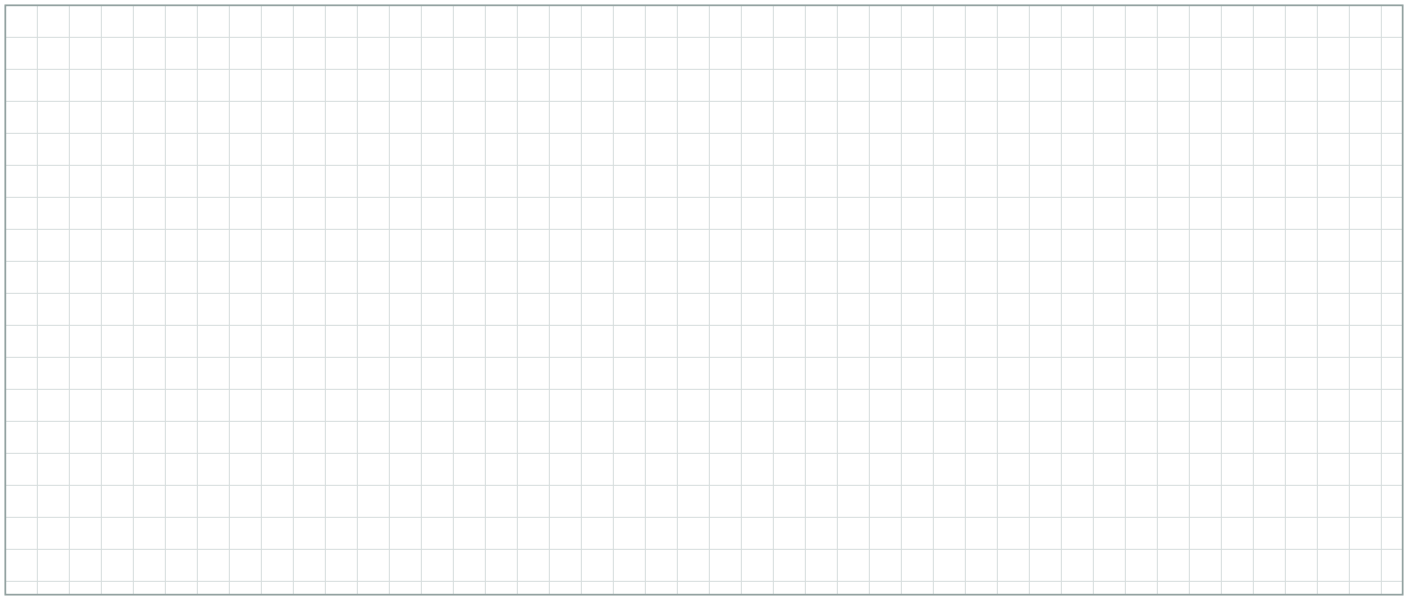
Choose an earlier artifact, a cured concrete sample comparison or an approved low-voltage device. Record the task, one changed factor and a fixed comparison method. For concrete, match age and dimensions. For a device, trace the low-voltage path.

Measure the current condition. Record the baseline, its units, and the test method before touching the artifact.

Test feature	Baseline and units	One change	Revised prediction
Primary measure			
Fixed condition			

02 Plan it on paper

Identify one change and predict the direction of its effect. For a concrete extension (2.6), plan batches and full cure before comparing samples.



My prediction, chosen feature or design limit

Repair and repeat a fair test

Flexible clinic / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Make the chosen repair or conduct a teacher-approved concrete or low-voltage consumer-device investigation. Record the changed variable and the result in this packet. Repeat the same test. Record a result that confirms or contradicts the prediction and discuss what to try next.

Trial	Before	After	Conditions or anomaly
1			
2			
3			

04 Use the reading

Ch. 14 | From the Ground Up | printed pp. 123-131
Ch. 23 | Down the Pipes | printed pp. 209-218
Ch. 30 | Electrifying! | printed pp. 271-280
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Which piece of evidence changed your decision? Cite the task, measurements, and what you will do next.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

PROJECT 4 / MEETING 11 / COMMUNICATION AND CONTROL

Demonstrate and defend a design

Exhibition / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Sketch the exhibit label with a named need, labeled mechanism, result and revision. Plan speaking roles. Choose a feature a visitor can locate or operate.

Recheck one dimension or operating value in front of a peer. Write the unit and the acceptance criterion.

Claim	Evidence and units	Who demonstrates it?
Need / criterion		
Mechanism		
Measured change		

02 Plan it on paper

Plan a short demonstration that shows the need, mechanism, measured result, and revision.

My prediction, chosen feature or design limit

Demonstrate and defend a design

Exhibition / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Demonstrate the device or model. Invite a peer to use the drawing to find a feature or operate one state. Rotate roles. Ask the reviewer to challenge one assumption. Demonstrate a repeat trial or explain what additional evidence is needed.

Visitor check	What happened	Response or next test
Find a feature		
Repeat trial		
Challenge assumption		

04 Use the reading

Ch. 9 | Like Nature Intended | printed pp. 71-79
Ch. 10 | Redesigning America | printed pp. 83-87
Ch. 20 | An Ingenious Engine | printed pp. 185-190
Ch. 30 | Electrifying! | printed pp. 271-280
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Independently defend a change using a reading idea, a drawing feature, and comparable results from your own tests.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.